applications beyond their traditional medical origins. A voxel (volume pixel) represents the 3D equivalent of a pixel, defining a discrete unit of volume within a 3D space. Unlike traditional geometric models that define surfaces and boundaries, voxel models partition space into a regular grid where each cell contains specific properties or material characteristics. This difference in representation makes voxels particularly suited for modeling heterogeneous materials, internal structures, and phenomena that vary continuously through volume rather than existing only at surfaces.

2.5 Spatial mapping and reference systems

We can see that digitally representing physical assets and objects as 2D representations, 3D models, and digitized data from documents is a powerful aid to understanding, but there is another dimension to this representation, and that is the position of the objects in the real world. Spatial reference systems are systems that allow us to precisely measure locations on the surface of the earth and with many different types in use, it is crucial to understand how these systems work to create an accurate spatial representation of reality where we can precisely position objects in virtual space and determine spatial relationships between objects.

2.5.1 Coordinate systems and projections

When modelling physical objects, there are a number of categories of reference systems that you need to be familiar with. Often, you will need to transform between each of these different systems when moving between representations, and it is important to understand what each type of reference system is, what it is used for, and how you may convert between each type.